

Geometry-Conditioned Turn Scoring for Conversational Memory Compression

Pranav Kompally

Northeastern University
United States of America
kompally.p@northeastern.edu

Sibi Chakkaravarthy S

VIT-AP University
India
chakkaravarthy.sibi@vitap.ac.in

Abstract

Compressing a long multi-turn conversation requires choosing which earlier turns to keep, compress, or remove. We study a sparse keep/compress/drop (KCD) codec and vary the signal used to rank candidate turns: hidden-state geometry, semantic similarity, and query-conditioned geometry within a semantic shortlist. On saved MSC runs (1,000 conversations), LongLLMLingua has larger raw next-token logit ℓ_2 differences than recency baselines while its teacher-forced continuation loss is lower. In a separate KCD-variant run, query-conditioned geometry has the lowest reported raw logit ℓ_2 and teacher-forced loss at budget 0.50. On a fixed 100-instance LongMemEval-S artifact, semantic KCD has the lowest teacher-forced loss among the reported policies. These findings are descriptive: raw logit distance is a model-state diagnostic, not an answer-quality metric, and the LongMemEval experiment does not run the official generated-answer evaluator. We document the codec, run configuration, budget accounting, and artifacts needed to reproduce or extend the evaluation.

1 Introduction

Long conversations create a memory-allocation problem before they create a generation problem. Once the full dialogue no longer fits inside a practical inference budget, a system must decide which parts of the conversation to retain verbatim, which parts to compress into sparse surrogates, and which parts to evict. This is not only a capacity problem: even when the full context fits, language models fail to uniformly exploit information distributed across long histories—performance degrades substantially for content appearing in the middle of the context window (Liu et al., 2024).

Existing approaches include retrieval augmentation (Lewis et al., 2020), external or hierarchical memory management (Wang et al., 2023; Zhong

et al., 2024; Packer et al., 2023; Wang et al., 2024), prompt compression (Jiang et al., 2023, 2024), and KV-cache compression or eviction (Liu et al., 2023; Zhang et al., 2023; Li et al., 2024; Cai et al., 2024; Kim et al., 2025). All are useful, but they motivate a central question for *in-context* conversational memory compression: *what signal should determine turn importance under a budget?* We focus on inference-time compression that operates within the context window without external indexes or model modification.

This paper studies that question through a sparse keep/compress/drop (KCD) codec for conversational memory. The codec is deliberately simple. Conversation segments can be kept in full, compressed into anchor-plus-support summaries, or evicted. The variable of interest is the *scoring signal* that drives those actions. We compare three regimes: geometry-based scoring from the conversation’s hidden-state trajectory, semantic scoring from query similarity, and a hybrid policy that uses geometry only *inside* a semantic shortlist.

The paper’s central claim is deliberately narrow. A turn-ranking signal can change the diagnostic and teacher-forced continuation metrics obtained by an otherwise fixed codec. We do not treat raw logit distance as task accuracy, claim a universally best signal, or infer a causal explanation for differences across models or benchmark lengths.

We first report a small exploratory stress-set pilot, then report saved MSC baseline and KCD-variant runs and a fixed LongMemEval-S diagnostic artifact. The paper distinguishes these runs rather than treating their point estimates as directly comparable across configurations.

This paper makes three main contributions:

- We specify a turn-structured KCD codec and three ranking signals with explicit state, cost, and tie-handling definitions.
- We report diagnostic raw-logit and teacher-

forced continuation-loss results from separate saved MSC and LongMemEval-S runs, with their different evaluation limitations made explicit.

- We provide the implementation and source-backed hard-set artifacts in the accompanying repository, and record the configuration and provenance limits of the saved public-benchmark outputs.

2 Related Work

Memory architectures and compression.

LongMem, MemoryBank, and MemGPT provide long-term or hierarchical memory mechanisms for language-model interaction (Wang et al., 2023; Zhong et al., 2024; Packer et al., 2023); MEMORYLLM instead incorporates a fixed-size, self-updatable latent memory pool (Wang et al., 2024). These systems expand or maintain memory capacity; our setting fixes the representation form and asks which signal allocates a limited turn budget. LLMLingua and LongLLMLingua perform prompt compression using token-level importance estimates (Jiang et al., 2023, 2024). COMEDY studies compressive memory for long-term conversations, R³Mem uses reversible compression for memory retention and retrieval, and COMEM studies compressed context memory for online language-model interaction (Chen et al., 2025; Wang et al., 2025; Kim et al., 2024). Our main distinction is to hold the codec fixed and vary the ranking signal.

Budgeted inference and token-level retention.

Scissorhands, H2O, SnapKV, PyramidKV, and EpiCache manage or compress KV caches using token- or head/layer-level retention rules (Liu et al., 2023; Zhang et al., 2023; Li et al., 2024; Cai et al., 2024; Kim et al., 2025). They target inference memory and are not directly token-budget-matched to our text-level turn actions. Our setting is turn-structured and evaluates raw next-token logit distance alongside teacher-forced continuation loss.

Benchmarks. MSC provides multi-session open-domain conversations where partners accumulate personal information (Xu et al., 2022). LongMemEval evaluates long-term interactive assistant memory, including retrieval and update demands (Wu et al., 2025). MemBench evaluates memory capability across factual and reflective

settings, while TReMu focuses on temporal reasoning in multi-session dialogue agents (Tan et al., 2025; Ge et al., 2025). These benchmarks motivate a narrow question: how do alternative signals rank turns in the saved configurations studied here?

3 Setup

3.1 Conversation-State Geometry

For a conversation prefix $x_{1:T}$, let

$$z_t = f_\theta(x_{1:t}) \in \mathbb{R}^d \quad (1)$$

be the final-layer hidden state at the final token of turn t after applying the model chat template to the prefix. We normalize onto the unit sphere:

$$h_t = \frac{z_t}{\|z_t\|} \in \mathbb{S}^{d-1}. \quad (2)$$

Within a segment S_j , we move one-step increments into a common tangent space using the logarithm map and parallel transport:

$$u_t = \text{PT}_{h_t \rightarrow q_j}(\log_{h_t}(h_{t+1})) \in T_{q_j}\mathbb{S}^{d-1}, \quad (3)$$

where q_j is a segment reference point taken as the Fréchet mean of the trajectory (Afsari, 2011). This allows local low-rank analysis and curvature-like instability scores to be measured in a common frame.

For a segment, q_j is computed by a fixed-iteration Fréchet-mean routine on its normalized states. The implementation clips the inverse-cosine argument to $[-1, 1]$, uses an $\epsilon = 10^{-8}$ norm guard, and treats a zero tangent vector as zero under transport. These operations provide a deterministic geometric feature construction; they do not establish that a raw geometric distance is shift-invariant or a proxy for answer correctness.

3.2 Query-Conditioned Geometry

For semantic-memory tasks, the geometry signal is not used directly. The semantic score is cosine similarity between normalized final-token states of the target and candidate turns. We retain the top $2k$ semantic candidates, where k is the number of turns that can be retained under the nominal budget, and compute geometry in the target turn’s local frame. For target state h_q , let

$$v_q^{(j)} = \log_{q_j}(h_q) \quad (4)$$

be the query direction in the same tangent space as the segment motion. We then derive query-conditioned segment features including projected curvature, projected subspace energy, and query-alignment magnitude. These features are converted into turn-level refinement scores only *inside* the semantic shortlist.

This makes the hybrid policy a refinement method, not an independent retrieval method: a turn excluded by the semantic shortlist cannot be recovered by the geometric stage. Accordingly, comparisons involving the hybrid policy test whether local trajectory information changes the ranking of semantically plausible candidates; they do not test geometry as a replacement for semantic candidate generation.

3.3 The Sparse KCD Codec

For each segment j , the codec selects an action

$$a_j \in \{\text{keep}, \text{compress}, \text{evict}\} \quad (5)$$

under a fixed budget. A compressed segment retains its anchor and selected support turns; a kept segment retains all of its original turns; an evicted segment retains none. Turn-only policies rank candidates by score density and use stable original-turn order to break ties, then greedily fill the token budget. Segment KCD policies instead solve the discrete keep/compress/evict allocation by dynamic programming under the same nominal budget. The compression operator itself is fixed within each run. What changes is the scoring signal:

- **geometry_KCD**: geometric risk from transported motion.
- **semantic_KCD**: semantic similarity to the target turn.
- **budget_aware_semantic_KCD**: semantic scoring with budget-aware span control.
- **semantic_query_conditioned_geometry_KCD**: semantic shortlist first, then query-conditioned geometry within that shortlist.

3.4 Budgeted Objective and Distortion Metrics

Given a target turn t^* and a compression policy π , we evaluate a compressed history by comparing it with the full-history next-token logit vector. Let

$\ell(x_{1:t^*})$ denote that vector under the full history and $\ell(\hat{x}_{1:t^*})$ the vector under compression. We use

$$\Delta_{\ell_2} = \|\ell(\hat{x}_{1:t^*}) - \ell(x_{1:t^*})\|_2 \quad (6)$$

as a raw model-state diagnostic. It is not invariant to logit shifts, and we do not interpret it as answer quality. The companion teacher-forced continuation metric is

$$\Delta_{\text{NLL}} = -\log p_{\theta}(y^* | \hat{x}_{1:t^*}) + \log p_{\theta}(y^* | x_{1:t^*}) \quad (7)$$

as the change in negative log likelihood for the observed assistant continuation y^* . Negative Δ_{NLL} means lower teacher-forced loss than the full-context reference. It is not a free-generation or official benchmark answer score.

The memory-control problem can be written as

$$\min_{\pi} \mathbb{E}[D(\pi)] \quad \text{s.t.} \quad \mathbb{E}[M(\pi)] \leq B, \quad (8)$$

where $D(\pi)$ is a diagnostic, $M(\pi)$ is memory cost, and B is the nominal historical-turn budget. In the saved LongMemEval-S artifact, prompt formatting and an uncompressed recent window make mean realized full-prompt token fractions range from 0.932 to 0.978 despite mean nominal historical fractions ranging from 0.177 to 0.479. We therefore do not present those rows as token-matched comparisons.

3.5 Memory Regimes

The experiments distinguish two memory regimes.

- **Constraint-critical memory**: exact user instructions, retrieval packets, code constraints, or compact fact bundles whose loss directly breaks downstream answers.
- **Semantic continuity memory**: persona statements, preferences, identity, and episodic facts whose value is largely determined by their relation to the current query.

The paper’s hypothesis is that these regimes favor different scoring signals inside the same codec.

4 Experimental Design

Benchmarks and models. We report: (1) a **nine-conversation exploratory hard-set policy pilot**, spanning three synthetic memory families and three compact models; (2) a separate **36-case Qwen2.5-0.5B support-retention analysis** over 12 cases per family; (3) two separate saved **MSC**

Table 1: Support-turn retention in the separate 36-case Qwen2.5-0.5B hard-set analysis at budget 0.35. Counts are descriptive case counts for geometry versus uniform.

Outcome	Cases
Geometry retains more support turns	14
Uniform retains more support turns	5
Tie on support-turn retention	17
Geometry retains the latest support turn when uniform drops it	13

valid runs over 1,000 conversations (Xu et al., 2022), one for external baselines and one for KCD variants; and (4) a fixed **LongMemEval-S** artifact of 100 full histories (Wu et al., 2025). Primary public-benchmark rows use Qwen2.5-1.5B-Instruct (qwen25_15b). The saved LongMemEval output contains 20 shard-ID files, partial rows, and a merge manifest; the fixed 100-instance subset is not claimed to be representative of the full benchmark.

Hard-set analyses. The nine-conversation pilot provides the three-model budget curves in Figure 1. Its separate 36-case analysis is restricted to Qwen2.5-0.5B and compares geometry with uniform support-turn retention at budget 0.35. Table 1 and the Figure 1 count panel report the same analysis. It is not the sample behind the nine-conversation curve summaries or a public-benchmark estimate.

Evaluation and statistics. We report raw next-token logit ℓ_2 and teacher-forced Δ_{NLL} , both averaged by conversation. The legacy reports include uncorrected Monte Carlo signflip values; we omit them because a finite 5,000-draw procedure cannot support an exact zero p-value and the planned comparisons were not multiplicity corrected. Thus, tables are descriptive point estimates, not confirmatory tests. New runs should use paired conversation-level bootstrap intervals, add-one Monte Carlo p-values, and Holm correction over pre-specified policy comparisons.

5 Constraint-Critical Memory: Why Signal Choice Matters

5.1 External Baselines on the Stress Set

The nine-conversation hard-set pilot also compares several non-KCD baselines. These point estimates are included as exploratory mechanism

Table 2: Exploratory external-baseline comparison in the nine-conversation hard-set pilot (qwen25_15b, 540 target-turn rows). Values are descriptive means of raw logit ℓ_2 and teacher-forced Δ_{NLL} ; lower is better within each metric.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	387.4	338.0	268.5	1.627	1.272	1.353
Recency	348.4	326.3	274.6	1.620	1.115	1.027
Recency-KCD	348.4	311.8	290.9	1.620	1.115	0.959
Lexical TF-IDF	381.9	323.1	276.9	1.553	1.004	0.801
LONGLLMLINGUA	385.6	372.4	369.8	1.252	0.758	0.697

Table 3: Exploratory geometry-aware control versus uniform retention in the nine-conversation hard-set pilot. Negative deltas favor geometry-aware retention; no hypothesis-test claims are made.

Model	$\Delta\text{Logit}@0.20$	$\Delta\text{Logit}@0.35$	$\Delta\text{NLL}@0.35$
qwen25_05b	-65.654	-99.503	-0.091
qwen25_15b	-33.647	-21.669	-0.181
smollm2_17b	-4.105	-11.329	-0.002

evidence, not as a public-benchmark or token-matched comparison.

In this pilot, LongLLMLingua has the smallest teacher-forced loss at all three budgets, while Recency or Recency-KCD has the smallest raw logit distance. For example, at $B=0.50$, LongLLMLingua’s raw logit distance is 369.8 versus 268.5 for uniform, while its teacher-forced loss is 0.697 versus 1.353 for uniform. The metrics therefore produce different orderings in this controlled set. This is a diagnostic observation rather than evidence about generated-answer accuracy.

5.2 Geometry-Aware Control Under Scarcity

We next compare geometry-aware retention with uniform retention in the same nine-conversation pilot. Negative deltas indicate smaller values under geometry-aware retention.

All three displayed model configurations have lower raw logit distance under geometry-aware retention at the two reported budgets. This limited pilot does not establish an architecture-wide effect or a generated-answer benefit, but it motivates examining the controlled signal swap below.

5.3 Signal-Swap Ablation

geometry_KCD and semantic_KCD use the same codec and differ only in their ranking signal. This is the pilot’s direct signal-control comparison.

At 0.20 and 0.35, the semantic policy has larger values on both reported diagnostics. At 0.50, raw logit distance is effectively tied in this small pilot while the teacher-forced loss difference remains

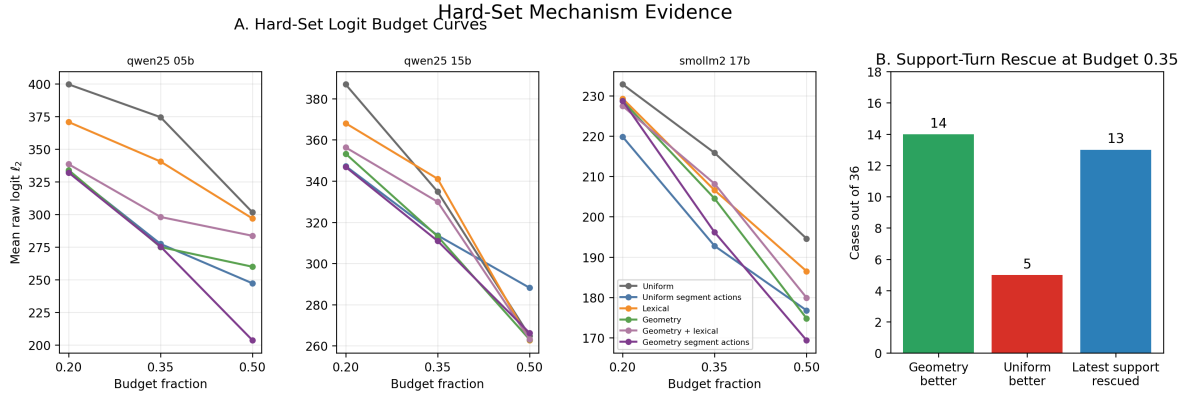


Figure 1: Hard-set diagnostic visualization from two distinct analyses. The left panel plots mean raw next-token logit ℓ_2 across budgets for the nine-conversation, three-model pilot. The right panel summarizes the separate 36-case Qwen2.5-0.5B support-retention analysis at budget 0.35: geometry retains more support turns in 14 cases, uniform in 5 cases, and geometry alone preserves the latest support turn in 13 cases. These are descriptive counts from the released per-case retention analysis.

Table 4: Exploratory signal-swap ablation in the nine-conversation hard-set pilot (qwen25_15b). Positive Δ means the semantic policy has a larger value than geometry_KCD.

Budget	$\Delta\Delta_{\ell_2}$	$\Delta\Delta_{\text{NLL}}$
0.20	+37.8	+0.195
0.35	+38.5	+0.424
0.50	+0.1	+0.628

positive. The separate 36-case support-retention analysis in Table 1 and Figure 1 is the paper’s mechanism evidence; it must not be numerically conflated with this nine-conversation signal-swap pilot.

6 Public Benchmarks: Diagnostic Results and Signal Variants

6.1 Geometry-Behavior Divergence: KCD vs. External Baselines

A central question for any compression method is whether behavioral quality metrics tell the full story. We compare KCD against LongLLMLingua (Jiang et al., 2024), recency-based, and lexical TF-IDF compression on MSC (1,000 conversations, 163,755 evaluation rows).

Within this saved run, LongLLMLingua has lower teacher-forced loss than the other listed policies at budgets 0.20 and 0.35, while its raw logit distance is 103–161 units above uniform. Relative to Recency-KCD, its raw-logit gaps are 125.5, 150.1, and 192.8; relative to ordinary Recency, they are 125.1, 168.4, and 219.7. The two diagnos-

Table 5: Saved MSC external-baseline run (qwen25_15b, 1,000 conversations). Values are conversation-level means of raw logit ℓ_2 and teacher-forced Δ_{NLL} ; they are descriptive and should not be compared with Table 6, which comes from a separate run configuration. Lower is better within each metric.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	627.2	613.8	596.3	3.361	3.269	3.171
Recency	605.3	572.6	537.5	3.236	3.050	2.901
Recency-KCD	604.9	590.9	564.4	3.219	3.021	2.860
Lexical TF-IDF	617.4	610.5	594.0	3.249	3.074	2.930
LongLLMLingua	730.4	741.0	757.2	3.114	3.070	3.013

tics therefore rank policies differently. This observation motivates reporting both, but it does not establish an effect on generated-answer correctness.

Recency-KCD has the smallest raw logit distance at budget 0.20, while ordinary Recency is smallest at budgets 0.35 and 0.50. The results are not token-matched with LongLLMLingua, so they should not be read as a fairness-controlled superiority claim.

6.2 MSC: KCD-Variant Run

Table 6 reports a separate saved MSC KCD-variant run. The comparison fixes the KCD family but is descriptive: the legacy statistical outputs are not used for confirmation.

At $B=0.50$, `sem_query_cond_KCD` has the smallest reported raw logit distance (705.5) and teacher-forced loss (2.671) in this run. Table 7 records component-removal point estimates. Removing support-score weighting increases both reported diagnostics at $B=0.50$, whereas removing query projection changes the two diagnostics differently. These are hypotheses for a pre-registered rerun

Table 6: Saved MSC KCD-variant run (qwen25_15b, 1,000 conversations). Values are conversation-level means of raw logit ℓ_2 and teacher-forced Δ_{NLL} . Lower is better within each metric; no significance claims are made.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	799.6	783.0	761.6	3.153	3.061	2.959
Geometry-KCD	804.9	814.5	812.8	3.105	2.973	2.811
Semantic-KCD	806.9	814.2	815.0	3.040	2.884	2.716
sem_query_cond_KCD	807.5	801.8	705.5	3.034	2.850	2.671

Table 7: Component ablation from the saved MSC KCD-variant run (qwen25_15b, 1,000 conversations). Values are descriptive conversation-level means; lower is better within each metric.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
sem_query_cond_KCD	807.5	801.8	705.5	3.034	2.850	2.671
w/o query proj.	806.7	809.4	708.3	3.028	2.856	2.670
w/o support score	808.9	817.0	757.7	3.094	2.957	2.762

rather than evidence for a causal mechanism.

6.3 LongMemEval-S: Fixed Teacher-Forced Diagnostic Artifact

LongMemEval-S rows come from a fixed artifact of 100 full histories. The experiment uses the observed assistant continuation for teacher forcing and does not run LongMemEval’s official generated-answer evaluator. It is consequently not an answer-accuracy or behavioral-generalization test.

Semantic-KCD has the lowest reported teacher-forced loss at each listed budget in this artifact. The ordering differs from the MSC KCD-variant run, but the differing dataset, prompt construction, and non-binding realized token budget prevent a causal or length-based explanation. The artifact is retained to make this non-replication visible rather than to support a benchmark-wide claim.

7 Cross-Model Descriptive Check

The submitted study also contains a 50-conversation MSC check using Llama-3.2-3B-Instruct. It keeps the policy family fixed but changes both model family and scale relative to Qwen2.5-1.5B; it is therefore a descriptive scope check rather than an architecture ablation.

Semantic-KCD has the smaller displayed values in this saved Llama run. This differs from the Qwen MSC KCD-variant ordering at budget 0.50, but one Llama configuration cannot identify whether the difference is due to model family, scale, tokenizer, post-training, or another run-specific factor. The result limits any model-

Table 8: Fixed LongMemEval-S teacher-forced diagnostic artifact (qwen25_15b, 100 full histories). Values are descriptive conversation-level means of raw logit ℓ_2 and teacher-forced Δ_{NLL} . The nominal budget is not equivalent to realized full-prompt token cost; no generated-answer or significance claim is made.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	1750.7	1781.8	1714.2	0.222	0.191	0.141
Geometry-KCD	1763.0	1701.3	1683.4	0.135	0.109	0.096
Semantic-KCD	1750.6	1700.5	1717.9	0.100	0.096	0.094
sem_query_cond_KCD	1755.9	1748.1	1705.0	0.108	0.105	0.099

Table 9: Descriptive 50-conversation MSC cross-model check (Llama-3.2-3B-Instruct). Values are conversation-level means of raw logit ℓ_2 and teacher-forced Δ_{NLL} ; lower is better within each metric.

Policy	L2@.20	L2@.35	L2@.50	NLL@.20	NLL@.35	NLL@.50
Uniform	689.7	669.4	631.0	3.722	3.657	3.524
Geometry-KCD	690.6	643.2	592.2	3.617	3.416	3.260
Semantic-KCD	671.9	633.5	586.1	3.526	3.317	3.139

general claim; it does not establish an explanation for the reversal.

8 Discussion and Limitations

The saved runs show that raw next-token logit distance and teacher-forced continuation loss can order compression policies differently. This is a useful diagnostic observation, not evidence that either quantity measures user-visible answer quality. The two measures answer different questions: teacher-forced loss evaluates probability assigned to one observed continuation, whereas raw logit ℓ_2 compares the entire next-token vector and is sensitive to shifts in that vector. Their agreement or disagreement is therefore a property to measure, not a reason to treat either metric as a universal objective. Centered logit distance, Jensen–Shannon divergence, and top- k overlap are implemented for future runs but are not reported here because they were not available for all legacy artifacts.

The paper has four principal limitations. First, it does not include LongMemEval’s official generated-answer evaluation, so it makes no LongMemEval answer-quality claim. Second, the 100-instance LongMemEval artifact is a fixed subset, not a documented random or representative sample. Third, its nominal historical-turn budgets do not match realized full-prompt token costs, precluding fair efficiency comparisons with external methods. Fourth, the reported point estimates come from separate saved configurations and legacy uncorrected significance reports; they should be reproduced with pre-specified compar-

isons, paired conversation-level confidence intervals, and multiplicity control. The primary model is Qwen2.5-1.5B-Instruct, a causal Transformer with RoPE, SwiGLU, RMSNorm, grouped-query attention, and tied word embeddings (Qwen Team, 2024). The reported results therefore support only this Qwen-family configuration; they do not identify an effect of any one architectural component.

8.1 Future Work

The next evaluation should run LongMemEval’s official generated-answer protocol on all 500 questions, with a fixed dataset revision, seed, and task-family breakdown. It should compare policies at matched realized input-token counts and report compression latency, peak memory, and preprocessing cost alongside quality. To test the diagnostic interpretation, it should report centered-logit distances, distributional divergences, and top- k agreement in addition to raw logit ℓ_2 . Finally, it should pre-specify primary contrasts, use paired conversation-level intervals with multiplicity correction, and evaluate several matched decoder-only model families. Such a study can test whether the observed Qwen result replicates across families or tracks tokenizer, post-training, scale, and architecture; the present experiments cannot distinguish those possibilities.

Acknowledgments

Experiments were conducted on cloud GPU infrastructure. This work received no external funding. The authors declare no conflicts of interest. The authors used Claude (Anthropic) to assist with manuscript drafting and editing; the authors reviewed and approved the final text.

References

- Bijan Afsari. 2011. Riemannian L^p center of mass: Existence, uniqueness, and convexity. *Proceedings of the American Mathematical Society*, 139(2):655–673.
- Zefan Cai, Yichi Zhang, Bofei Gao, Yuliang Liu, Tianyu Liu, Keming Lu, Wayne Xiong, Yue Dong, Baobao Chang, Junjie Hu, and Wen Xiao. 2024. [Pyramidkv: Dynamic kv cache compression based on pyramidal information funneling](#). *arXiv preprint arXiv:2406.02069*.
- Nuo Chen, Hongguang Li, Jianhui Chang, Juhua Huang, Baoyuan Wang, and Jia Li. 2025. [Compress to impress: Unleashing the potential of compressive memory in real-world long-term conversations](#). In *Proceedings of the 31st International Conference on Computational Linguistics*, pages 755–773.
- Yu Ge, Salvatore Romeo, Jinghui Cai, Raphael Shu, Yassine Benajiba, Monica Sunkara, and Yi Zhang. 2025. [Tremu: Towards neuro-symbolic temporal reasoning for llm-agents with memory in multi-session dialogues](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 18974–18988.
- Huiqiang Jiang, Qianhui Wu, Chin-Yew Lin, Yujiu Yang, and Lili Qiu. 2023. [Llmllingua: Compressing prompts for accelerated inference of large language models](#). In *Proceedings of the 2023 Conference on Empirical Methods in Natural Language Processing*, pages 13358–13376.
- Huiqiang Jiang, Qianhui Wu, Xufang Luo, Dongsheng Li, Chin-Yew Lin, Yuqing Yang, and Lili Qiu. 2024. [Longllmllingua: Accelerating and enhancing llms in long context scenarios via prompt compression](#). In *Proceedings of the 62nd Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 1658–1677.
- Jongwon Kim, Sungdong Kim, Sanghwan Bae, and Minjoon Seo. 2024. [COMEM: Compressed context memory for online language model interaction](#). In *The Twelfth International Conference on Learning Representations*.
- Minsoo Kim, Arnav Kundu, Han-Byul Kim, Richa Dixit, and Minsik Cho. 2025. [Epicache: Episodic kv cache management for long-term conversation on resource-constrained environments](#). *arXiv preprint arXiv:2509.17396*.
- Patrick Lewis, Ethan Perez, Aleksandra Piktus, Fabio Petroni, Vladimir Karpukhin, Naman Goyal, Heinrich Küttler, Mike Lewis, Wen-tau Yih, Tim Rocktäschel, Sebastian Riedel, and Douwe Kiela. 2020. Retrieval-augmented generation for knowledge-intensive NLP tasks. In *Advances in Neural Information Processing Systems*, volume 33, pages 9459–9474.
- Yuhong Li, Yingbing Huang, Bowen Yang, Bharat Venkitesh, Acyr Locatelli, Hanchen Ye, Tianle Cai, Patrick Lewis, and Deming Chen. 2024. [Snapkv: LLM knows what you are looking for before generation](#). In *Advances in Neural Information Processing Systems*, volume 37, pages 22947–22970.
- Nelson F. Liu, Kevin Lin, John Hewitt, Ashwin Paranjape, Michele Bevilacqua, Fabio Petroni, and Percy Liang. 2024. [Lost in the middle: How language models use long contexts](#). *Transactions of the Association for Computational Linguistics*, 12:157–173.
- Zichang Liu, Aditya Desai, Fangshuo Liao, Weitao Wang, Victor Xie, Zhaozhao Xu, Anastasios Kyrilidis, and Anshumali Shrivastava. 2023. [Scissorhands: Exploiting the persistence of importance hypothesis for llm kv cache compression at test time](#). *arXiv preprint arXiv:2305.17118*.

- Charles Packer, Sarah Wooders, Kevin Lin, Vivian Fang, Shishir G. Patil, Ion Stoica, and Joseph E. Gonzalez. 2023. [Memgpt: Towards llms as operating systems](#). *arXiv preprint arXiv:2310.08560*.
- Qwen Team. 2024. [Qwen2.5: A party of foundation models](#).
- Haoran Tan, Zeyu Zhang, Chen Ma, Xu Chen, Quanyu Dai, and Zhenhua Dong. 2025. [Membench: Towards more comprehensive evaluation on the memory of llm-based agents](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 19336–19352.
- Weizhi Wang, Li Dong, Hao Cheng, Xiaodong Liu, Xifeng Yan, Jianfeng Gao, and Furu Wei. 2023. [Augmenting language models with long-term memory](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Xiaoqiang Wang, Suyuchen Wang, Yun Zhu, and Bang Liu. 2025. [R³Mem: Bridging memory retention and retrieval via reversible compression](#). In *Findings of the Association for Computational Linguistics: ACL 2025*, pages 4541–4557.
- Yu Wang, Yifan Gao, Xiusi Chen, Haoming Jiang, Shiyang Li, Jingfeng Yang, Qingyu Yin, Zheng Li, Xian Li, Bing Yin, Jingbo Shang, and Julian McAuley. 2024. [Memoryllm: Towards self-updatable large language models](#). In *Proceedings of the 41st International Conference on Machine Learning*, volume 235 of *Proceedings of Machine Learning Research*, pages 50453–50466.
- Di Wu, Hongwei Wang, Wenhao Yu, Yuwei Zhang, Kai-Wei Chang, and Dong Yu. 2025. [Longmemeval: Benchmarking chat assistants on long-term interactive memory](#). In *International Conference on Learning Representations*.
- Jing Xu, Arthur Szlam, and Jason Weston. 2022. [Beyond goldfish memory: Long-term open-domain conversation](#). In *Proceedings of the 60th Annual Meeting of the Association for Computational Linguistics (Volume 1: Long Papers)*, pages 5180–5197.
- Zhenyu Zhang, Ying Sheng, Tianyi Zhou, Tianlong Chen, Lianmin Zheng, Ruisi Cai, Zhao Song, Yuandong Tian, Christopher Ré, Clark Barrett, Zhangyang Wang, and Beidi Chen. 2023. [H2o: Heavy-hitter oracle for efficient generative inference of large language models](#). In *Advances in Neural Information Processing Systems*, volume 36.
- Wanjun Zhong, Liang Guo, Qiqi Gao, Han Ye, and Yanlin Wang. 2024. [Memorybank: Enhancing large language models with long-term memory](#). In *Proceedings of the AAAI Conference on Artificial Intelligence*, volume 38, pages 19191–19199.

A Implementation and Artifact Details

The implementation and source-backed figure-generation utilities are available in the accompanying repository: <https://github.com/SteveMama/rt-geometry-memory>. The main text specifies the state token and layer, semantic score, shortlist, cost unit, actions, allocation procedures, and tie handling. This appendix records the run lineages and limitations needed to interpret the displayed descriptive results. The extraction cache is keyed by model, conversation identifier, and extraction settings; queue jobs are durable across restarts and write per-conversation partial rows. Candidate prompts are deduplicated by retained-turn indices within a target turn before batched scoring. These optimizations change wall-clock cost, not the evaluated prompt or policy.

B Evaluation Provenance

The MSC baseline and KCD-variant tables are distinct saved runs and retain their own output directories. The LongMemEval-S artifact comprises 100 fixed identifiers, 20 shard-ID files, per-shard partial CSV files, and a merge manifest. Its rows use teacher forcing over observed continuations. The artifact does not contain official generated answers, generated-answer scores, or a token-matched comparison protocol. The hard-set pilot and Llama check are reported as descriptive evidence with their sample identities stated explicitly. The separate Qwen2.5-3B hard-set validation remains excluded because its source artifacts still need reconciliation before publication.

C Additional Exploratory Diagnostics

Geometry characterization. In the submitted characterization artifact, the estimated rank-95 summaries for the three compact-model trajectories were 1.049, 1.167, and 1.528; their turn-order-shuffled counterparts were 1.486, 1.458, and 1.799. The corresponding reported geometry–

decoder-drift correlations were 0.989, 0.994, and 0.989. These values motivate testing transported geometry as a control feature, but they are not used as evidence that raw logit distance predicts user-visible answer quality.

Negative extensions. Several extensions did not improve the reported KCD policies. A two-head harm predictor ranked held-out turn-removal rows reasonably (held-out Kendall $\tau = 0.612$) but was poorly calibrated across conversations, so it was not used as a final policy. Marker-based object-level compression underused its target budget, realizing only 43–52% of the nominal budget in the failing policies. Finally, exploratory update-detection variants based on transported-increment sign structure did not reliably distinguish updates from controls. These negatives further bound the present claim: the evidence concerns turn ranking within the tested KCD codec, not a general-purpose memory-state detector or learned harm model.

Object-level checkpoint. The saved MSC checkpoint below makes the budget-underuse issue concrete. It is a diagnostic result from a separate configuration, not a token-matched comparison with the main tables.

Table 10: Exploratory object-level checkpoint on MSC valid (qwen25_15b, nominal budget 0.35). Lower is better for the two diagnostics. The object policies realize materially less than the target budget.

Policy	Raw logit ℓ_2	Teacher-forced Δ_{NLL}	Realized token fraction
Semantic	928.9	2.438	0.743
Budget-aware semantic KCD	931.2	2.487	0.736
Semantic object KCD	970.9	2.979	0.519
Object harm KCD	1023.0	3.251	0.431

The marker-based boundaries over-segmented the conversations and the within-object compressor then retained too little context. This is why these policies are treated as failed exploratory variants, rather than evidence about the value of object-level memory in general.